Remote Sensing of Environment 247 (2020) 111946

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

DeepCropMapping: A multi-temporal deep learning approach with improved spatial generalizability for dynamic corn and soybean mapping

Jinfan Xu[a], Yue Zhu[a], Renhai Zhong[a], Zhixian Lin[a], Jialu Xu[a], Hao Jiang[a], Jingfeng Huang[b], Haifeng Li[c,*], Tao Lin[a,*]

[a] College of Biosystems Engineering and Food Science, Zhejiang University, Zhejiang, Hangzhou 310058, China
[b] Institute of Applied Remote Sensing and Information Technology, Zhejiang University, Hangzhou, China
[c] School of Geosciences and Info Physics, Central South University, Changsha, Hunan, China

ARTICLE INFO

Keywords:
Crop mapping
Model generalization
Remote sensing
Time series
Long short-term memory
Deep learning

ABSTRACT

Accurate crop mapping provides important and timely information for decision support on the estimation of crop production at large scale. Most existing crop-specific cover products based on remote sensing data and machine learning algorithms cannot serve large agriculture production areas as a result of poor model transfer capabilities. Developing a generalizable crop classification model for spatial transfer across regions is greatly needed. A deep learning approach, named DeepCropMapping (DCM), has been developed based on long short-term memory structure with attention mechanisms through integrating multi-temporal and multi-spectral remote sensing data for large-scale dynamic corn and soybean mapping. Full cross validation of classification experiments were conducted in six sites each covering 2,890,000 pixels at 30 m resolution in the U.S. corn belt from Year 2015 to 2018. Landsat Analysis Ready Data (ARD) and Cropland Data Layer (CDL) were adopted as the input satellite observations and ground reference, respectively. Transformer, Random Forest (RF), and Multilayer Perceptron (MLP) models were built for comparison. The DCM model produced a mean kappa score of 85.8% in base sites and a mean average kappa score of 82.0% in transfer sites at the end of the growing season. It yielded a comparable performance to Transformer and better than RF and MLP at the local test. The DCM model significantly outperformed other three models with a 95% confidence interval in the spatial transfer analysis. The results demonstrated the capability of learning generalizable features by the DCM model from ARD time series. The computational complexity analysis suggested that the DCM model required a shorter training time than Transformer but longer than MLP and RF. The results of the in-season classification experiment indicated the DCM model captured critical information from key growth phases and achieved higher accuracy than other models after the beginning of July. By monitoring the classification confidence in each time step, the results showed that the increased length of seasonal remote sensing time series would reduce the classification uncertainty in all sites. This study provided a viable option toward large-scale dynamic crop mapping through the integration of deep learning and remote sensing time series.

1. Introduction

Trusty and timely crop mapping is essential for monitoring dynamic crop growth and assessing agricultural resources toward food security (Thenkabail et al., 2012). Accurate early-season crop classification provides important information for decision support systems of government and agriculture-related applications, including crop insurance, land management, and supply-chain logistics (Cai et al., 2018). It is especially critical for multiple agents dealing with extreme weather events derived from global climate change occurs in the early stage of the growing season (Nelson et al., 2009; Vermeulen et al., 2012). To understand cropping pattern changes and estimate potential socio-economic impacts, reliable near-real-time crop mapping is greatly needed. With the advent of an increased number of remote sensing data, satellite-based methods aided by machine learning algorithms become the predominant approach of land cover mapping and produce crop type maps with various spatial resolutions and scales (Azzari and Lobell, 2017).

As publicly available label databases for constructing classifiers cover a limited spatial extent, detailed maps toward agriculture land use are highly localized. Many agriculture production areas lack reliable medium-resolution crop-specific ground references for directly

* Corresponding authors.
E-mail addresses: lihaifeng@csu.edu.cn (H. Li), lintao1@zju.edu.cn (T. Lin).

https://doi.org/10.1016/j.rse.2020.111946
Received 16 March 2020; Received in revised form 7 June 2020; Accepted 8 June 2020
Available online 26 June 2020
0034-4257/ © 2020 Elsevier Inc. All rights reserved.

training supervised learning models. The scarcity of crop type labels has become one of the major challenges in generating large-scale field-level crop map products, although the spatial and temporal resolutions of satellite products gradually improve with the launch of new earth-observing satellites (Wang et al., 2019). The collection of ground annotations heavily depends on field surveys and censuses, which are financially expensive, labor-intensive, and time-consuming. A few studies labeled remote sensing images manually by visual comparison with high-resolution RGB photos from GPS records and Google Earth. These methods, however, require great extra efforts from experts and are not feasible for large-scale monitoring programs (Dong et al., 2015; Zhang et al., 2015). Two typical national-scale medium-resolution regularly-updated crop type maps are Cropland Data Layer (CDL) published by the United States Department of Agriculture (USDA) and Crop Inventory (CI) published by the Agriculture and Agri-Food Canada (AAFC). They provide annually-updated crop type maps with resolutions of 30 m based on substantial labeled training samples thanks to comprehensive field surveys (AAFC, 2018; USDA, 2018). Such capability of collecting timely ground annotations and producing accurate crop type maps are not readily available in other countries (Song et al., 2017).

A viable solution is transferring well-trained classification models from a local region with rich ground reference data to target regions. Few studies have explored the spatial generalizability of crop and natural vegetation mapper, and most of them were dependent on external ancillary data to varying degrees (Cohen et al., 1998; Pax-Lenney et al., 2001; Wang et al., 2019; Woodcock et al., 2001; Zhong et al., 2019b). Building a spatially generalizable crop classification model remains a scientific challenge due to the spatial heterogeneity of meteorological conditions, hybrid selection, topographical features, and farming practice (Edreira et al., 2014; Lobell and Azzari, 2017; Zipper et al., 2016).

Previous studies have demonstrated that employing phenological profiles to characterize crop growth can enhance model generalizability across years and regions (Wang et al., 2019; Zhong et al., 2014). It provides an opportunity to transfer classification models across regions with similar crop phenology, despite the spatial heterogeneity of the above geographical factors. Learning efficient feature representation to characterize inherent temporal cumulative growth of crops is critical for improving model generalization. Developing a multi-temporal approach for dynamic crop mapping base on the time series analysis of remote sensing observations is greatly needed.

Multi-temporal satellite observations have been fed into machine learning models including Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM) and Multilayer Perceptron (MLP) (Khatami et al., 2016; King et al., 2017; Löw et al., 2013; Massey et al., 2017; Shi and Yang, 2016). These models are originally designed to process non-temporal high-dimensional data instead of multi-temporal remote sensing data. The sequential relationship hidden in time-series profiles is not explicitly utilized and thus handcrafted and pre-defined temporal features are incorporated into input variables collections. Statistical values of time series including mean, median, standard deviation and amplitudes of spectral bands and vegetation indices were adopted for discriminating between 18 agriculture-related land cover types in France (Pelletier et al., 2016). Key transition dates and corresponding values derived from vegetation index curves such as global start and peak points have been widely used and improved classification performance (Brown et al., 2010; Simonneaux et al., 2008; Simonneaux and Francois, 2003; Walker et al., 2015). Many studies have employed phenological metrics to represent crop characteristics. These approaches leveraged curve-fitting methods such as wavelet transformation (Sakamoto et al., 2006), spline fitting (Bradley and Mustard, 2008) and double logistic functions (Soudani et al., 2008), and fed major function parameters and aggregated information in each phases into classifiers (Pelletier et al., 2016; Siachalou et al., 2015; Zhong et al., 2014). Those empirical feature extraction, however, heavily rely on domain knowledge and expertise. Feature engineering is a time-consuming and challenging task due to the complex variation of hybrids, meteorological conditions and light-scattering mechanisms. Considering a large number of phenological metrics are unavailable until the end of the growing season, they are not suitable for producing in-season crop maps.

Recent advances in deep learning have demonstrated that end-to-end neural network approaches can discover intricate relationships from high-dimensional data (LeCun et al., 2015). Convolutional Neural Network (CNN), Recurrent Neural Network (RNN) and self-attention network with positional encoding are three prominent deep learning architectures that can efficiently process sequential data. CNN has been adopted to extract multilevel features in spatial and spectral domains from remote sensing imagery (Huang et al., 2018; Kussul et al., 2017; Marcos et al., 2018), but rarely applied in the temporal domain for time series analysis (Zhong et al., 2019a, 2019b). RNN is specialized for learning sequential relationship by explicitly linking adjacent observations (Werbos, 1990). RNN performs well in dealing with sequential data such as text data in natural language modeling (Sundermeyer et al., 2012) and acoustic data in speech processing (Sak et al., 2014). In the field of land cover mapping, some studies have leveraged RNN to recognize temporal features from multi-temporal remote sensing data (Jia et al., 2017; Lyu et al., 2016; Mou et al., 2019). Self-attention structures were proposed for sequence analysis in natural language processing in 2017 (Vaswani et al., 2017). It supports feature extraction from a very long sequence which can exceed 10,000 units with efficient parallel computing. The transformer model, which combining multiple self-attention layers with shortcut connection and layer normalization, has achieved state-of-the-art performance in many natural language processing applications (Dai et al., 2019; Devlin et al., 2019). Although it has not been fully explored in remote sensing communities, the latest studies have suggested its competitive performance in crop type classification (Garnot et al., 2019; Rußwurm and Körner, 2019).

Long Short-Term Memory (LSTM) model is a variant of vanilla RNN, which facilitates time-series analysis and allows exhibiting long-term temporal dependencies by gated mechanisms (Hochreiter and Schmidhuber, 1997). The multi-temporal architecture of LSTM naturally suits crop type identification by modeling vegetation cycles from long observation sequences. LSTM has been applied to classifying multi-class crop types in recent studies and showed the effectiveness of capturing dynamic temporal characteristics (Rußwurm and Körner, 2017; Zhong et al., 2019a). To further improve the temporal feature extraction of straightforward LSTM models, attention mechanisms and bi-directional flows have been introduced and yielded state-of-the-art classification performance for crop type identification (He et al., 2019; Rußwurm and Körner, 2018). The potential of LSTM in crop mapping under various situations including spatial transfer and in-season classification has not been fully tested. The uncertainty and reliability of LSTM-derived maps are also relatively unexplored.

The objective of this study is to assess the spatial generalizability of the LSTM-based multi-temporal approach for end-of-the-season and in-season crop mapping. The in-season crop type maps are valuable and rarely publicly available, given that annual CDL maps are released several months after the end of the growing season. In this paper, we developed an attention-based bidirectional long short-term memory model named DeepCropMapping (DCM) to classify corn and soybean in six sites in the U.S. corn belt. One site was selected as the base site for model training and other five sites were used for the performance test of spatial generalizability. We used historical data (2015–2017) from the base site to build models and applied the models to crop mapping for the next year (2018) in all sites. Landsat Analysis Ready Data (ARD) and Cropland Data Layer (CDL) were adopted as the input satellite observations and ground references, respectively, for producing 30 m resolution maps. Transformer, Random Forest (RF) and Multilayer Perceptron (MLP) were built for comparison. End-of-the-season and in-

season experiments were conducted to evaluate model performance at different stages of the growing season. To further explore the reliability of DCM, an uncertainty analysis was performed by monitoring classification confidence over time, which provides an improved understanding of the learning procedure of DCM. This study aims to address three following research questions:

(RQ1) How well does DCM generalize spatially for end-of-the-season crop mapping compared with Transformer, RF, and MLP?
(RQ2) How does DCM perform for in-season crop mapping compared with Transformer, RF, and MLP?
(RQ3) How does DCM learn generalizable features from time-series profiles as the growing season progresses?

2. Materials and methods

2.1. Classification approach

2.1.1. DeepCropMapping

The DeepCropMapping (DCM) model receives a pixel-level time-series of satellite observations during the crop growing season and yields category probabilities as the soft output and a predicted category as the hard output. The model employs multiple LSTM layers with bi-directional flows and attention mechanisms to understand complicated temporal contexts and learn long-term temporal patterns from satellite time-series data. Specifically, the model contains three components: input module, Attention-based Bidirectional Long Short-Term Memory (AtBiLSTM) module, and output module (Fig. 1).

2.1.1.1. Input module. Each pixel-level ARD observation at time step t is encoded as a vector $[sb_1, sb_2, \ldots, sb_S]$ called x_t. It consists of multiple spectral bands, where S is the number of bands. The input is expressed as a time sequence $[x_1, x_2, \ldots, x_t, \ldots, x_T]$, where T is the length of observation sequence fed into the network.

2.1.1.2. AtBiLSTM module. In this study, two LSTM layers extract high-level abstract temporal features hierarchically from the input sequence. Each LSTM layer is composed of LSTM units, in which information is selectively stored at the current time step t and selectively transported to the next time step $t + 1$ to construct the complex connections between time steps. Compared to vanilla RNN, the main advantage of LSTM units for handling sequence data is leveraging the adaptive gating mechanism to determine the degree to which LSTM units keep the previous state and memorize the current features (Zhou et al., 2016). Based on the gating mechanism, LSTM units generates the hidden feature vector $h_t^{forward}$ at time step t as Eqs. (1)–(5) demonstrate:

$$g_t^f = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

$$g_t^i = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

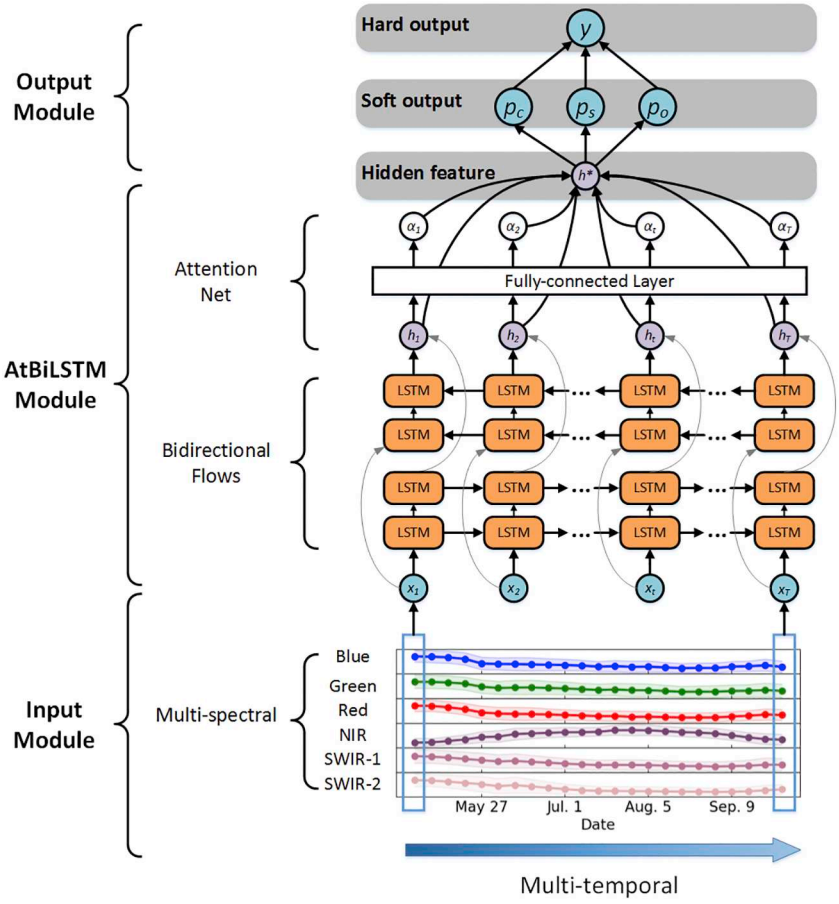


Fig. 1. The architecture of DeepCropMapping (DCM) for multi-temporal and multi-spectral crop mapping.

$$S_t^{forward} = g_t^f \times S_{t-1}^{forward} + g_t^i \times tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (3)$$

$$g_t^o = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (4)$$

$$h_t^{forward} = g_t^o \times tanh(S_t^{forward}) \quad (5)$$

where W_f, W_i, W_c, W_o, b_f, b_i, b_c and b_o are learnable weight matrices and σ and *tanh* are activation functions. $S_t^{forward}$ is the cell state vector and represents the information selectively accumulated from time step 1 to time step t. Three typical gates are: forget gate vector g_t^f, which decides how much information from the previous time step to keep; input gate vector g_t^i, which decides how much information from the current input to take; and output gate vector g_t^o, which decides how much information from the current time step to output. To facilitate filtering information, each number in gate vectors ranges from zero to one and represents the trade-off of removing and reserving. LSTM layers transfer the hidden feature vector $h_t^{forward}$ and the cell state vector $S_t^{forward}$ from the start to the end of time series step by step and thus learn temporal features in the forward direction. In addition to the forward flow, bidirectional LSTM layers build a backward flow simultaneously and produce $h_t^{backward}$ with the same shape as $h_t^{forward}$. LSTM units within each flow share all weight matrices and do not interact across flows. The bidirectional flows increase network capacity and provide comprehensive temporal context for one time step. The hidden feature vectors from two opposite directions are connected to obtain the final output hidden feature $[h_t^{forward}, h_t^{backward}]$ called h_t. The dimension of hidden feature h_t is determined after testing the candidate values of 128, 256, 512, and 1024. Bidirectional LSTM layers process the whole time series and output hidden feature matrix $[h_1, h_2, \ldots, h_t, \ldots h_T]$ from the last LSTM layer. To better aggregate the information from multiple time steps for crop type classification, the attention module adjusts the contribution of each hidden feature to the output module through normalized weight parameters. The attention net, composed of a one-layer fully-connected network with the Softmax function, produces a weight value α_t for each hidden feature vector h_t obtained from the bidirectional LSTM module. For each time step t, the adjusted hidden feature h_t^* is calculated by Eqs. (6) and (7):

$$\alpha_t = softmax(W_{attention} \cdot h_t + b_{attention}) \quad (6)$$

$$h_t^* = \alpha_t h_t \quad (7)$$

Where $W_{attention}$ and $b_{attention}$ are learnable weight matrices in the attention net and shared by all time steps. *softmax* is used to normalize the weight value α_t, so that the sum of all weight values is equal to one. All adjusted hidden feature vectors are summed to generate the final representation of the whole time sequence. The output of the attention module is the aggregated hidden feature vector h^* equal to $h_1^* + h_2^* + \ldots + h_t^* + \ldots + h_T^*$.

2.1.1.3. Output module. This module generates soft output for uncertainty analysis and hard output for accuracy assessment. Single fully-connected layer with the Softmax function in the output module receives the hidden feature vector h^* as input and produces a discrete probability distribution consisting of three predicted probabilities corresponding to corn, soybean and "other" category. The probability distribution is used as the soft output of DCM. The predicted probability corresponding to the correct category is selected as classification confidence and used in uncertainty analysis. The category with the highest probability is adopted as the predicted category called the hard output.

The cross-entropy function is used as the loss function and the backpropagation algorithm with Adam optimizer (Kingma and Ba, 2014) is used for training the network. Dropout technique (Srivastava et al., 2014) and L2 norm regularization are applied to prevent the model from overfitting. Dropout randomly ignores some neurons in each LSTM layer during training and activate all neurons during test to simulate a special ensemble of multiple submodels built randomly. The probability of dropout is set to 50%. L2 regularization adds a penalty to the original loss to avoid too large weight parameters. The adjusted loss function is calculated as follows (Eq. (8)):

$$loss = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{K} p_{i,j}^{true} \log(p_{i,j}^{pred}) + \lambda \|w\|^2 \quad (8)$$

where N is the number of samples in the training set, i is the index of the sample, K is the number of categories, j is the index of the category, $p_{i,j}^{true}$ and $p_{i,j}^{pred}$ are the true and predicted probability that the ith sample belongs to the jth category, respectively, λ is the L2 regularization parameter and w is the weight matrix including all learnable parameters in DCM. The settings for constructing and training the DCM model are presented in Table S1.

2.1.2. Transformer

A Transformer model was developed based on self-attention mechanisms for crop type classification by adopting the encoder architecture of the original Transformer networks (Vaswani et al., 2017). The Transformer model consists of an input module, encoder module, and output module. The structures of input module and output module are similar to those of DCM (Fig. 1). A positional encoding tensor is added to the raw time series in the input module, as suggested by Vaswani et al. (2017). The encoder module implements feature extraction by stacking multiple multi-head self-attention layers each followed by one fully-connected layer. Shortcut connection and layer normalization operations are applied after each layer. The transformed features outputted by the last fully-connected layer are globally averaged through the time dimension and then fed into the output module to predict crop types. Detailed encoder architecture and multi-head self-attention mechanisms can be referred to the work by Rußwurm and Körner (2019).

To optimize the structure of the Transformer model, we searched the best configuration of the dimension of self-attention hidden feature in {64, 128, 256, 512}, the number of self-attention heads in {4, 6, 8}, the number of self-attention layers in {2, 4, 6}, and the dimension of fully-connect layer in {512, 1024, 2048}, as suggested by Rußwurm and Körner (2019) and Garnot et al. (2019). The settings of the loss function, optimizer, L2 norm regularization were the same as those used in the DCM model.

2.1.3. Baseline classification models

Two conventional classification models including Random Forest (RF) and Multilayer Perceptron (MLP) were developed for comparison. As an ensemble of decision-tree classifiers using the bagging strategy (Breiman, 1996), RF improves model generalizability by means of limiting the number of samples and features to consider when searching the best split. RF is well-known for avoiding overfitting and handling a large number of input variables with limited computation complexity (Breiman, 2001). RF is often established as baseline models since it has been one of the most successful ensemble learning methods in land cover mapping tasks (Belgiu and Drăguţ, 2016; Rodriguez-Galiano et al., 2012). MLP is designed for capturing complicated nonlinear features from high-dimensional data by stacking multiple fully-connected layers. MLP has been extensively applied in identifying crop types from satellite data (Cai et al., 2018; Carpenter et al., 1999; Dixon and Candade, 2008; Mas and Flores, 2008). The comparison between

MLP and LSTM would highlight the difference between the conventional structure and multi-temporal structure of neural networks, even if they both base on fundamental neuron connections. In this study, given an input time sequence x consisting of T time steps and S spectral bands, x was reshaped into one-dimensional vector x^* of length $S \times T$ and then fed into RF and MLP to predict crop types.

The best hyperparameter configuration was searched to optimize the structure of RF and MLP. For RF, the optimized hyperparameters include the number of trees called n_estimator and the number of features to consider when looking for the best split called max_features. In this study, n_estimator was determined after testing the candidate values of 100, 200, 500 and 1000 and max_features was determined after testing the candidate values of 2, 4, $\log_2(S \times T)$, $\sqrt{S \times T}$. For MLP, the number of hidden layers and the number of neurons in each hidden layer determine the depth and width of the network, respectively. The search range of the number of hidden layers was from 2 to 5. Correspondingly, the ratio of neuron number in difference layers was fixed at 1:1, 1:2:1, 1:2:2:1 or 1:2:4:2:1. The values of 128, 256, 512 and 1024 were tested for the neuron number of the first hidden layer. The settings of loss function, optimizer, L2 norm regularization and dropout were the same as those used in the DCM model.

2.2. Data

This study focused on the pixel-level mapping of corn and soybean in the U.S. corn belt. Corn and soybeans account for two of the top three types of land cover in most areas of the U.S. corn belt, and the rest are mostly non-cropland including grassland, pasture and deciduous forest, according to the 2018 CDL (USDA, 2018). As shown in Fig. 2, six sites were chosen to study the spatial transfer of crop classifiers, each covering 51 km × 51 km. These sites have a hot-summer humid continental climate with precipitation of about 1000 mm per year. Their crop production systems are mainly rain-fed, and the growing season of corn and soybean spans from late-April to mid-September and from mid-May to late-September, respectively. A single cropping system is typically practiced in these areas. As a result, we took corn, soybean and "other" (i.e., merged class of other land cover/land use types) as three classes of interest and assigned a label to each pixel of the thematic maps annually. The proportions of corn, soybean and "other" in six sites are reported in Table 1. Cropland covers most of the area in Sites A, B, C and D, while Sites E and F are mainly non-cropland. Spatial transfer across these sites provides an opportunity to evaluate the model performance on the dataset with various unbalanced land cover types. This study considered the changes in land cover patterns over

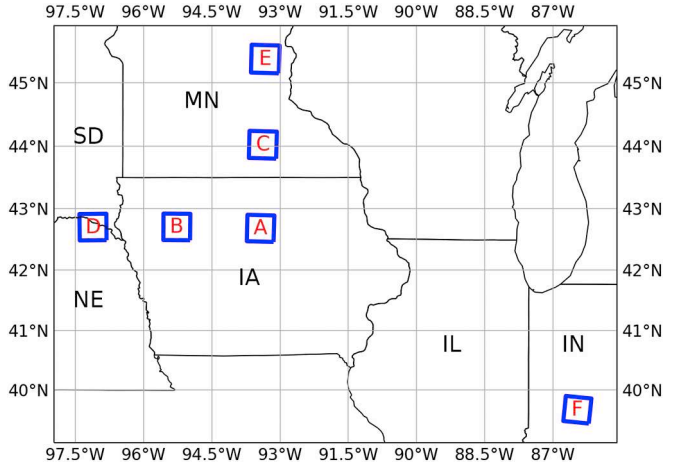


Fig. 2. Geography of six study sites (Sites A, B, C, D, E and F) in the U.S. corn belt. The state abbreviations are as follows: IA stands for Iowa, MN stands for Minnesota, SD stands for South Dakota, NE stands for Nebraska, IL stands for Illinois, and IN stands for Indiana.

Table 1
Summary statistics of valid pixels and crop-specific cover types in each site and each year.

Site	Year	Number of valid pixels	Validity percentage	Proportion of crop-specific cover types		
				Corn	Soybean	Other
Site A	2015	2,880,946	99.7%	51.6%	30.1%	18.3%
	2016	2,859,643	99.0%	55.8%	26.1%	18.1%
	2017	2,885,211	99.8%	52.0%	30.0%	17.9%
	2018	2,883,194	99.8%	51.6%	30.6%	17.8%
Site B	2015	2,833,040	98.0%	41.4%	37.4%	21.3%
	2016	2,854,056	98.8%	43.5%	35.4%	21.1%
	2017	2,739,248	94.8%	42.2%	36.9%	20.9%
	2018	2,611,813	90.4%	43.1%	37.6%	19.3%
Site C	2015	1,494,330	51.7%	40.1%	29.9%	30.0%
	2016	1,822,518	63.1%	43.9%	28.0%	28.1%
	2017	2,471,655	85.5%	40.7%	32.8%	26.4%
	2018	2,639,289	91.3%	41.2%	31.5%	27.3%
Site D	2015	2,594,266	89.8%	37.1%	26.2%	36.7%
	2016	2,762,543	95.6%	36.5%	26.3%	37.1%
	2017	1,962,044	67.9%	37.4%	28.5%	34.1%
	2018	2,712,551	93.9%	35.3%	29.3%	35.5%
Site E	2015	2,592,260	89.7%	7.8%	5.0%	87.2%
	2016	2,452,733	84.9%	7.5%	6.3%	86.2%
	2017	1,382,124	47.8%	6.5%	6.7%	86.9%
	2018	1,123,324	38.9%	7.3%	7.5%	85.2%
Site F	2015	1,699,073	58.8%	24.3%	23.1%	52.6%
	2016	2,740,183	94.8%	22.0%	19.3%	58.7%
	2017	2,668,841	92.4%	20.0%	22.9%	57.1%
	2018	2,844,635	98.4%	19.2%	22.0%	58.8%

years, which are caused by crop rotation and cropland and non-cropland conversion as reported in Table S2. Most corn and soybean farmers adopt crop rotation cycles in the study area. The cropland conversion are relatively limited except Site E, where more than 20% of cropland had been changed to non-croplands from 2015 to 2018.

Landsat Analysis Ready Data (ARD) surface reflectance composites from April 22 to September 23 were used as inputs for the crop growing season in each year, developed by the United States Geological Survey (USGS ARD, 2018) and publicly available from USGS's EarthExplorer web portal (https://earthexplorer.usgs.gov). Combining the Landsat 7 Enhanced Thematic Mapper Plus (ETM+) and Landsat 8 Operational Land Imager (OLI) products, ARD provides non-spatially overlapping imageries at 30 m resolution. Sites A, B, C, D, E and F (each of 2,890,000 (1700 × 1700) pixels) are fully encompassed in the footprint of h18v07, h17v07, h18v06, h16v07, h18v05 and h22v09 in the ARD grid system, respectively. The detailed information of Landsat ARD tile grid system can be found at https://www.usgs.gov/media/images/conterminous-us-landsat-analysis-ready-data-ard-tiles. Six optical bands including blue, green, red, near-infrared, shortwave infrared 1 and shortwave infrared 2 at each observation date were used as the input variables for the classification models. We discarded any pixel observations labeled cloudy, shadow, filled or unclear. Any pixel with less than seven valid observations after May 15 was considered to lack sufficient multi-temporal information and excluded from the dataset. The summary statistics of valid pixels in each site and each year are listed in Table 1. In order to remove the gaps resulted from missing acquisitions and invalid observations, linear temporal interpolation was conducted for imputing missing values, which has been proved to be effective in previous study (Inglada et al., 2015). Linear interpolation based on the nearest valid values before and after the target time step for each pixel was applied to obtain a time series consisting of 23 time steps with intervals of seven days. April 15 is assumed to be the earliest planting date in this study based on the state-level corn and soybean progress reports (USDA-NASS, 2018). The time steps are from week 1 (April 22) to week 23 (September 23) after April 15, which covers the crop progress from planting to mature stage of corn and soybean for these states (USDA-NASS, 2018). An example is shown in Fig. 3 for Site

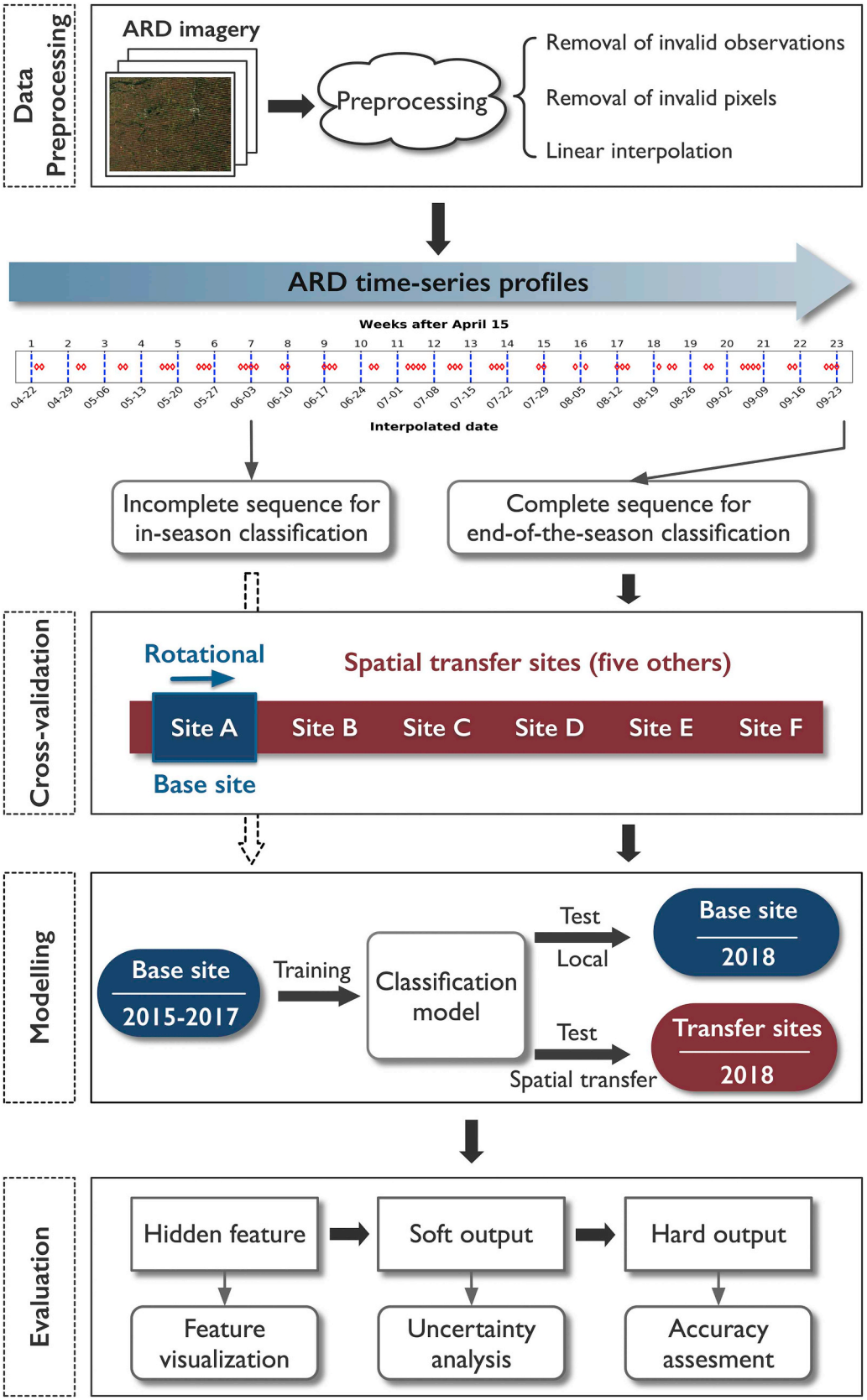


Fig. 3. Workflow of the end-of-season and in-season crop mapping experiment. The calendar of raw and interpolated ARD time series of Site A in 2018 is shown here to illustrate the data preprocessing for end-of-the-season and in-season classification experiments. In the illustration of ARD time-series profiles, the red rhombuses refer to raw acquisitions and the blue dashed lines refer to interpolated dates and corresponding weeks after April 15. It is worth noting that the raw acquisition here does not exclude the invalid pixel-level data and each pixel has an individual raw time series of valid observations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

A in 2018 on the calendar of raw and interpolated ARD time series.

USDA's CDL downloaded from the CropScape website portal (https://nassgeodata.gmu.edu/CropScape/) was used as the reference map for both training and test datasets. The 30 m CDL map was derived from a variety of Landsat Thematic Mapper imagery, Common Land Unit (CLU), NASS June Agriculture Survey and National Land Cover Dataset (NLCD) and generated in the Albers Equal Area Conic projection (Boryan et al., 2011). Providing more than 100 land cover and crop

type categories, CDL has a high total mapping accuracy for the major crop types, especially greater than 95% for corn and soybeans in the U.S. corn belt (Boryan et al., 2011). CDL was exploited as ground reference data in our experiments.

2.3. Experiment design

End-of-the-season and in-season classification experiments were designed to evaluate the classification capability of DCM (RQ 1 and 2). The workflow of our experiments is illustrated in Fig. 3. One site was chosen as the base site, and other five sites were regarded as the spatial transfer sites. Full cross-validation was conducted by rotating Sites A, B, C, D, E and F in turns for the base site in the end-of-the-season classification experiment. All models were trained only by the data from the base site. In order to mimic real-life situations, we used historical data to build models and applied the models to crop mapping for the next year. To be specific, we fed the data of the base site from 2015, 2016, and 2017 into DCM, Transformer, RF, and MLP for training and predicted the crop types of all sites in 2018. For instance, if we choose Site A as the base site, then the sample size of the training set is 8,625,800 and that of the test set is 14,814,806 in total. In the end-of-the-season classification experiment, models could utilize the complete time-series data spanning from April 22 to September 23. The in-season classification experiment, however, required models to extract useful information from the remote sensing time series with limited length. Models could only see the partially unmasked time series to predict crop types in this scenario. For example, the classification map at week 12 was generated by the classification model designed to receive a time series consisting of the first 12 weeks after April 15 as input. The length of input observation sequence gradually increased until all weeks were included. It is a simulation of the practical situation that more and more satellite images are available as the growing season progresses. The difference between the data structures employed by models in processing samples can be reflected when they are forced to perform classification without sufficient satellite observations. For the in-season classification experiment, Site A was fixed as the base site and other five sites were regarded as the spatial transfer sites without cross-validation.

The thematic uncertainty of in-season classification performance was investigated for a better understanding of the learning procedure of DCM (RQ 3). The model produced soft prediction for each sample in the form of probability by the Softmax function in the output module. We monitored the classification confidence, the predicted probability of the correct crop types, during the growing season to explore the learning procedure.

The modeling process was performed on a Linux workstation (Ubuntu 16.04 LTS) with two Intel Xeon Gold 6230 Processors (2.1G/20 Core/27.5 M), 128 GB of RAM, and six NVIDIA GeForce RTX 2080 Ti graphics cards (11 GB of RAM). All the models were implemented on the Python platform using the PyTorch library for DCM, Transformer, and MLP, as well as the Scikit-learn library for RF.

2.4. Model evaluation

Model performance was evaluated by comparing the classification results with CDL maps for the year 2018. Cohen's kappa coefficient, macro-averaged user's accuracy, and macro-averaged producer's accuracy were computed as overall performance indicators for all models. F1 score was used for a further performance measure for each land cover type. Cohen's kappa coefficient is calculated from the probability of observed agreement and expected agreement given by a hypothetical random classifier using Eqs. (9)–(11):

$$kappa = \frac{p_o - p_e}{1 - p_e} \tag{9}$$

$$p_o = \frac{1}{N}\sum_{i=1}^{K} n_i^{corr} \tag{10}$$

$$p_e = \frac{1}{N^2}\sum_{i=1}^{K} n_i^{pred} n_i^{true} \tag{11}$$

where *kappa* is the Cohen's kappa coefficient, p_o is the probability of observed agreement also known as overall accuracy, p_e is the probability of expected agreement given by a hypothetical random classifier, N is the number of samples, K is the number of categories, i is the index of the category, n_i^{corr} is the number of correctly classified samples in the ith category, n_i^{pred} and n_i^{true} are the number of samples in the ith predicted and reference category respectively. The macro-averaged user's accuracy (Eq. (12)) and macro-averaged producer's accuracy (Eq. (13)) are the simple average of user's accuracy and producer's accuracy for all classes, respectively, and the F1 score is the harmonic mean of the user's and producer's accuracy for each class (Eq. (14)):

$$Macro - averaged\ UA = \frac{1}{K}\sum_{i=1}^{K} UA_i \tag{12}$$

$$\text{Macro} - averaged\ PA = \frac{1}{K}\sum_{i=1}^{K} PA_i \tag{13}$$

$$F1_i = 2\frac{UA_i * PA_i}{UA_i + PA_i} \tag{14}$$

where $Macro - averaged\ UA$ is the macro-averaged user's accuracy, $Macro - averaged\ PA$ is the macro-averaged producer' accuracy, $F1$ is the F1 score, K is the number of categories, i is the index of the category, UA_i is the user's accuracy of the ith category and PA_i is the producer's accuracy of the ith category.

3. Results and discussion

3.1. Spatiotemporal analysis based on time series comparison

Discriminating between corn and soybean was challenging due to their similar ARD time-series profiles (Fig. 4(a)). The difficulties of crop mapping were demonstrated by aggregating spectral information over time of corn, soybean and “other” pixels. ARD time-series profiles of “other” category were clearly distinct from those of corn and soybean. For visible bands and shortwave infrared bands, “other” category had lower reflectance values at the beginning of the growing season around week 5 and higher valley values around week 17. The near-infrared bands of “other” category also showed a difference from crop types in initial values, peak values and gradients. The similar vegetation cycles of corn and soybean, however, made it difficult to recognize their differences in spectral reflectance curves. The buffer areas of visible bands overlapped throughout the growing season although soybean had higher mean reflectance values than corn. In the period from week 11 (July 1) to week 16 (August 5), corn and soybean curves had few overlaps in shortwave infrared bands. This period might provide critical features for corn and soybean mapping.

The separability between different land cover types varied across

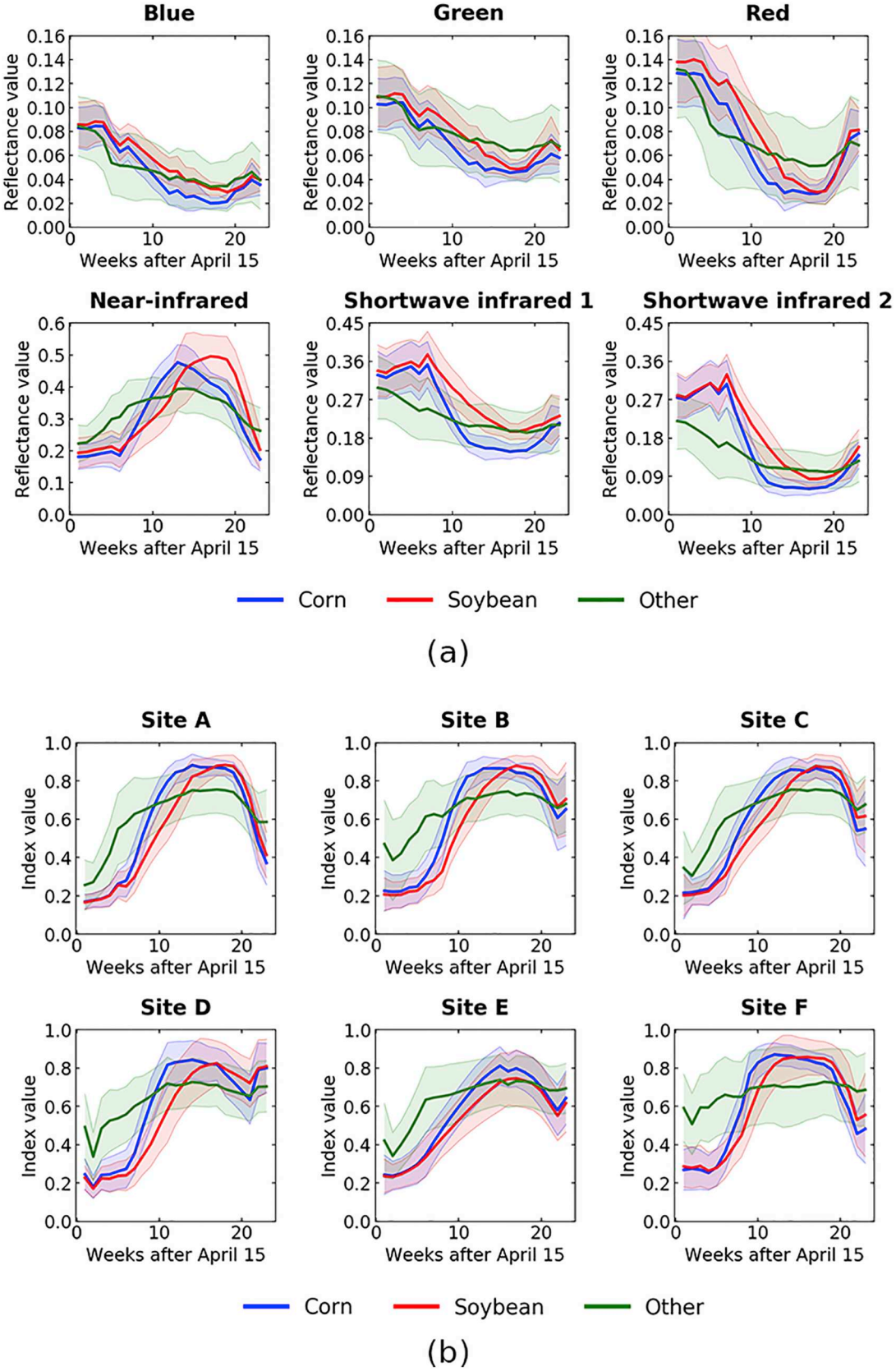


Fig. 4. Time-series analysis of ARD spectral reflectance of corn, soybean and "other" pixels in 2018. (a) All ARD spectral reflectance for Site A. (b) Temporal profiles of Normalized Difference Vegetation Index (NDVI) for all sites. The blue lines represent the average values of corn, the red lines represent the average values of soybean and the green lines represent the average values of "other". The light buffer areas refer to one standard deviation from the average value. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

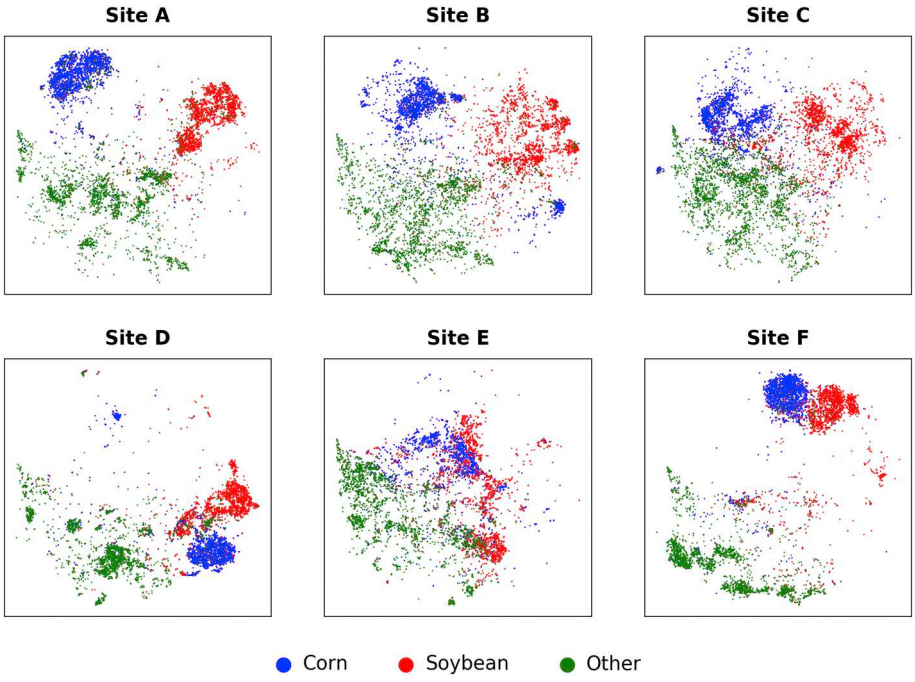


Fig. 5. Visualization of multi-spectral and multi-temporal ARD data by t-SNE for the year 2018. Samples in all sites and all classes are put together in a t-SNE visualization and grouping subplots by sites. Each point stands for one sample. Individual axes have no quantitative interpretation due to the non-linear dimensionality reduction. The clusters indicate that the samples with similar features have been detected and grouped together.

regions, which were reflected in the results of dimensionality reduction of ARD time-series profiles (Fig. 5). In addition to statistical information comparison among sites (Fig. 4(b)), we applied t-distributed stochastic neighbor embedding (t-SNE) (van der Maaten and Hinton, 2008) for qualitative measure of sample separability. For each land cover type and site, 3000 samples were randomly selected and nonlinearly projected to a two-dimensional space for visualization by t-SNE since spatial analysis in the original 138-dimensional feature space is challenging. The crop samples were generally better grouped than "other" category in all sites as shown in Fig. 5, which could be explained by the multiple land cover types, including deciduous forest, woody wetlands, and developed space, within "other" category. The features of corn and soybean samples in Sites A, B, C, D and F showed relatively high separability, while those in Site E were poorly separable. These results indicated that the boundary between the clusters of different land cover types varied across six study sites. Although the t-SNE visualization did not support distance based quantitative measure of sample separability, the difference of the extent of cluster formation suggested the varied difficulty in crop type classification across regions. Learning efficient and generalizable feature representation, especially for summer crops, remained a scientific challenge for all classifiers in spatial transfer tasks.

3.2. End-of-the-season classification

3.2.1. Performance analysis

The DCM and Transformer models always outperformed RF and MLP in local tests (Fig. 6(a)). The DCM model yielded a significantly higher kappa score than RF and MLP with a 95% confidence interval in local tests (Table 2). The DCM model produced a mean kappa score of 85.8%, which is superior to RF (83.9%) and MLP (83.9%). DCM showed similar improvement over RF and MLP in user's accuracy (Fig. S1(a)) and producer's accuracy (Fig. S2(a)), indicating its advantage in validity and completeness of prediction results. The Transformer model achieved comparable performance to DCM, which agreed with a previous study by Rußwurm and Körner (2019). The DCM and Transformer models always provided the best crop type map.

The DCM model produced improved spatial generalized results across sites than Transformer, MLP, and RF, with a significantly higher kappa score at a 95% confidence interval in spatial transfer sites (Table 2). The improvement of DCM over RF and that over MLP in average transfer kappa score reached the maximum value of 6.5% and 6.0% in Site F, respectively (Fig. 6(b)). The comparison in the user's accuracy (Fig. S1(b)) and producer's accuracy (Fig. S2(b)) similarly demonstrated the advantage of DCM. The Transformer model showed relatively unstable performance in different transfer sites. The average

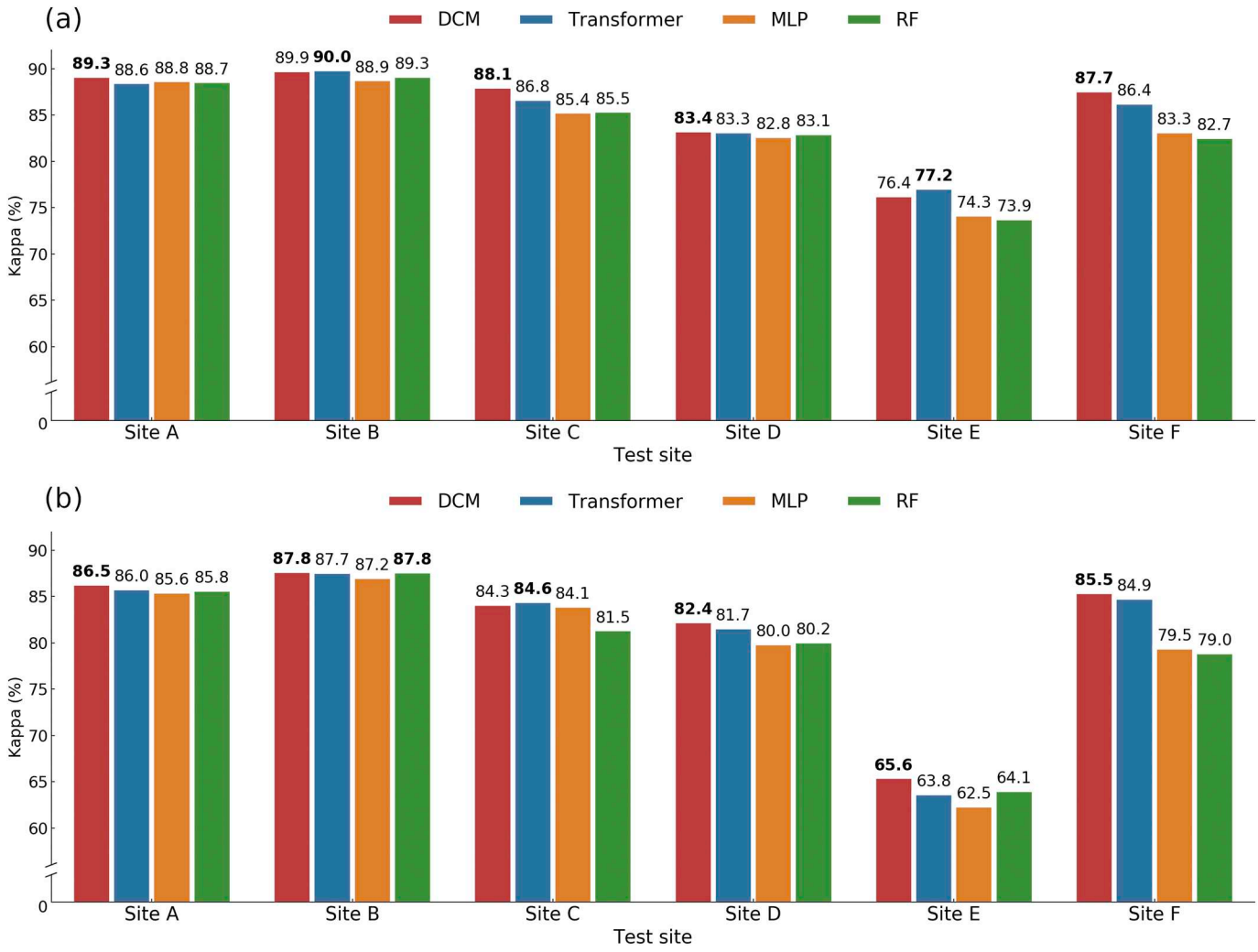


Fig. 6. Overall performance evaluation (Cohen's kappa coefficient) of DCM, Transformer, RF and MLP for six sites in the end-of-the-season classification experiment: (a) local test performance and (b) average transfer performance. All models are trained on the dataset from 2015 to 2017 and tested in 2018. The local test performance is obtained when the training and test process are conducted in the same site. The average transfer performance of a particular site is calculated by averaging the classification performance of the models trained in other sites and tested in this site. The bold values stand for the best score of four models.

Table 2
P values of paired two-sample *t*-test based on Cohen's kappa score for the comparison of DCM to Transformer, MLP, and RF in end-of-the-season classification experiments.

	Local test	Spatial transfer test
Transformer	0.285	< 0.05
MLP	< 0.05	< 0.05
RF	< 0.05	< 0.01

kappa score of Transformer was second to DCM in Sites A, D, and F, even slightly higher than DCM in Site C, but lower than RF in Sites B and E. The results demonstrated the strong and stable spatial generalizability of DCM in crop mapping tasks. The DCM model yielded the most competitive performance with a mean average transfer kappa score of 82.0%, as compared to its mean local kappa score of 85.8% (Fig. 6). The confusion matrices of the test set by the DCM model in the spatial transfer from Site A to other sites were provided in Fig. 7 as an example to directly report concrete prediction results on the dataset with various unbalanced land cover types. The crop type maps for each site, which was produced by the DCM model trained in Site A, were shown in Fig. 8 for visual comparison with CDL reference maps.

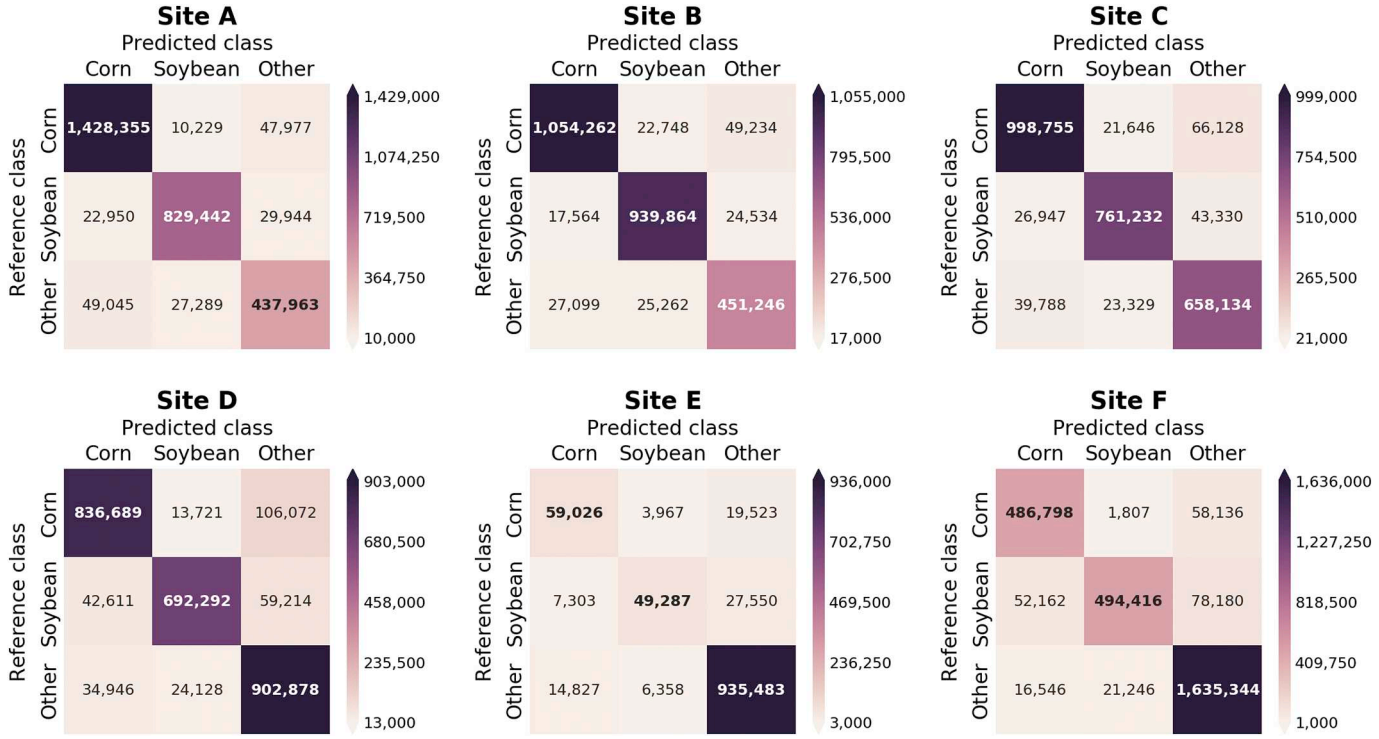


Fig. 7. Confusion matrices of the test set by the DCM model which are trained in Site A from 2015 to 2017 and tested in all sites for the year 2018 in the end-of-the-season classification experiment. Values in confusion matrices represent the number of samples. Diagonal values stand for the number of correctly classified samples.

All four models showed certain performance drop for transfer analysis as compared to local tests. The magnitude of performance decrease in different sites could be explained by the degree of similarity of phenological cycles indicated by ARD time-series profiles. Four models suffered a mean performance decrease of 11.5% in Site E, as compared to a decrease of less than 3.0% in other sites(Fig. 6). The distinct phenological characteristics of Site E might account for the result. The Normalized Difference Vegetation Index (NDVI) curves indicated the difference of the phases and amplitude of phenology between Sites E and other sites, though they had similar overall phenological cycles (Fig. 4(b)). The peak index value of soybean in Site E was relatively low. The date corresponding to the peak index value for soybean was earlier and for corn was later than most other sites. The seasonal tendency of NDVI curves of corn and soybean showed higher similarity in Site E, while the curve reached two separated peaks for corn than soybean in other sites. The low proportions of corn and soybean (Table 1) and the scatter of crop parcels (Fig. 8) in Site E might cause sample impurity such as mixed pixels, which could result in similar temporal profiles and lower the classification accuracy (Chen et al., 2016). Overall, such differences led to the performance dip since crop growth information extracted from other sites was not as effective for Site E as it was for local areas. Model transfer provided satisfactory performance in other sites. For example, the average transfer kappa score of the DCM models transferred from other sites to Site D was only 1.0% lower than the local kappa score in Site D (Fig. 6). The results showed the applicability of spatial transfer of DCM across the regions.

The success of DCM was more attributed to its accurate recognition of crop patterns than “other” category (Fig. 9). The DCM model outperformed the other three models with the highest average F1 score for all land cover types. The advantage of DCM and Transformer over RF and MLP was more significant for corn and soybean than “other” category, indicating the strength of these two deep learning models in discriminating between crop types. With the recurrent architecture, DCM modeled the crop growth progress to learn accumulated time-series spectral features for an improved generalized classification. It is interesting to find that all models achieved a higher F1 score for “other” category than crop types. Separating specific crop types was more challenging than discriminating between cropland and non-cropland, which was consistent with the analysis of Fig. 4(a).

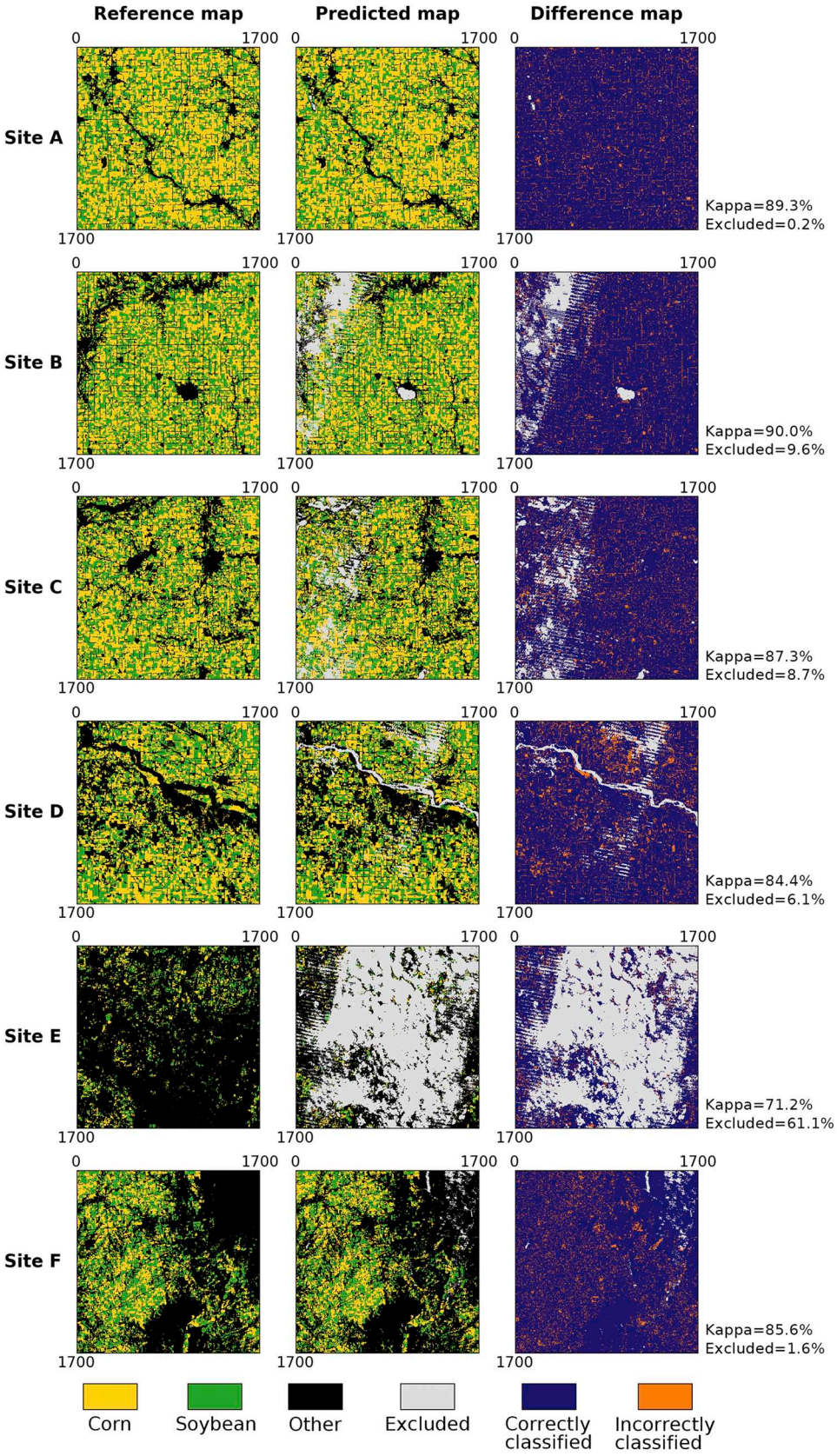


Fig. 8. The spatial distribution of crop classification results for the year 2018 in six sites. (a) CDL reference maps; (b) The crop type maps generated by the DCM model that is trained in Site A from 2015 to 2017; (c) The difference maps of DCM classification results compared with CDL ground reference. Correctly classified pixels are in dark blue while misclassified pixels are in orange. The pixels excluded from the dataset due to the lack of sufficient observations are in gray. The percentage of exclusion denoted by "Excluded", and kappa score denoted by "Kappa", are reported on the right side of each row. Each map contains 2,890,000 (1700 × 1700) pixels at 30 m resolution. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

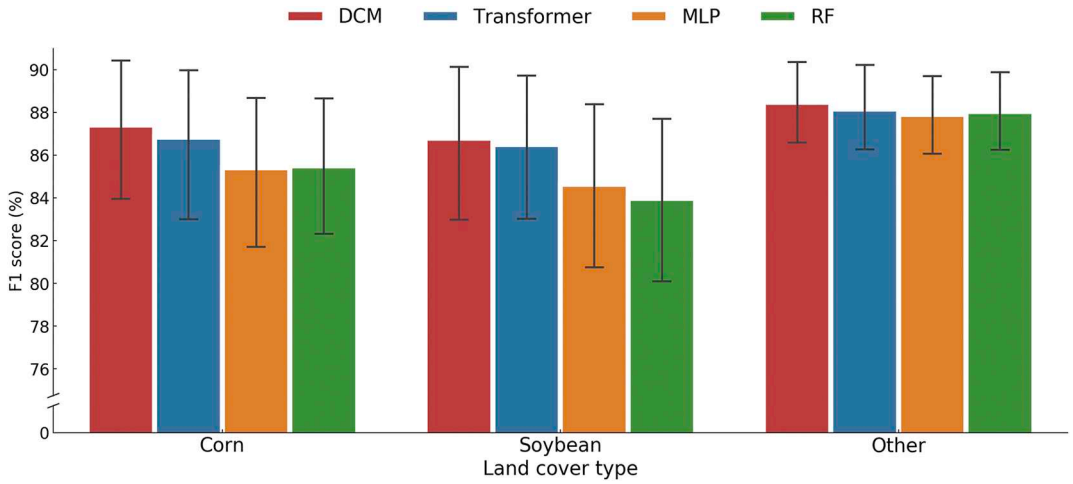


Fig. 9. F1 score of four models for three land cover types in all cases including local test and spatial transfer.

3.2.2. Computational complexity analysis

The DCM model required a shorter training time than Transformer but longer than MLP and RF in all sites in the end-of-the-season classification experiment (Fig. 10). The training process of the Transformer model cost up to 303.6 min on average, which was much longer than DCM (56.1 min), MLP (19.0 min), and RF (12.1 min). The DCM model contained an average of 4.5 M parameters, which was smaller than Transformer (18.9 M). The large number of parameters of the Transformer model resulted in high computational complexity and long computing time, although Transformer employed self-attention layers instead of recurrent network structure in its encoder module to make full use of the powerful parallel computing ability of graphics processing units (GPU). It is worthwhile mentioning that the RF model led to high memory usage in this study due to the large sample size. This may restrict its application in large-scale training tasks. All four models required a relatively short local test time of less than 1.5 min in all sites (Fig. S3).

The training time of a particular site was related to sample size and optimal hyperparameter settings. All models cost the shortest training time in Site C since it had the least training samples with a sample size of 5,788,503 while other sites had an average of 7,581,242 training samples. The datasets in different sites could result in different hyperparameter settings for optimal performance. For example, the DCM model contained 2.1 M and 8.4 M parameters due to different hyperparameter settings when trained in Sites A and B, respectively. The training time, therefore, showed a great difference in these two sites though they had a similar sample size.

3.3. In-season classification

The DCM model showed performance improvement over RF and MLP at the early stage of the growing season (Fig. 11). Site A was fixed as the base site and other five sites were regarded as the spatial transfer sites in the in-season classification experiment. The Transformer model yielded close but unstable performance as compared to DCM, especially in Site E where its kappa score could be higher at the beginning but lower than MLP in the later period from week 16 to 18. Although all models benefited from more inputs from satellite images, DCM gradually outperformed RF and MLP after the beginning of July, about week 12 (July 8). From that time, according to the temporal analysis of Fig. 4(a) in Section 3.1, the spectral reflectance curves with a buffer area of corn and soybean started to separate. The corn in the study area was approaching silking stage while the soybean was approaching blooming stage. The spectral information of corn and soybean began to show significant differences. The shortwave infrared profiles of corn and soybean had few overlaps. For the near-infrared band, the reflectance of corn was around the peak and that of soybean was still rapidly rising. DCM began to show performance advantages since it captured critical information from these key growth phases.

Spatial differences of ARD time-series profiles across sites affected the shape of kappa score curves (Fig. 11). The curve of Sites A, B and C was smoother than that of other sites. There were still some obvious fluctuations on the curve of Sites D, E and F, although in general the kappa scores for all sites improved with the increased length of the input series until they reached a plateau. It could be explained by the

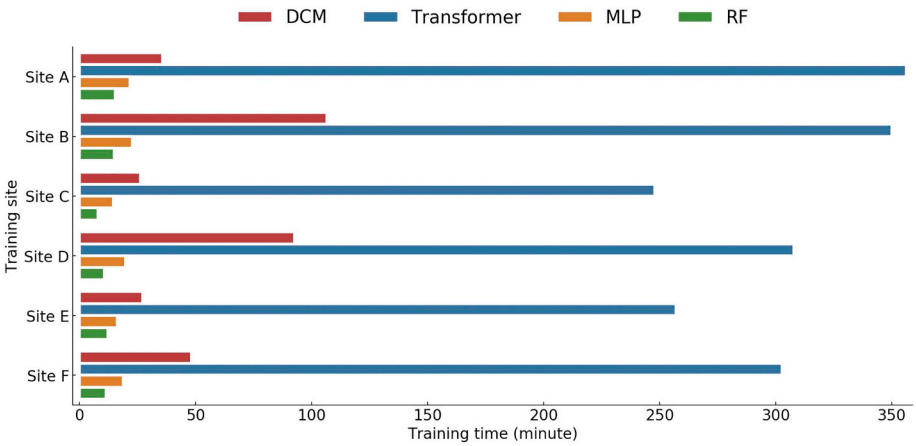


Fig. 10. Training time of four models in six sites.

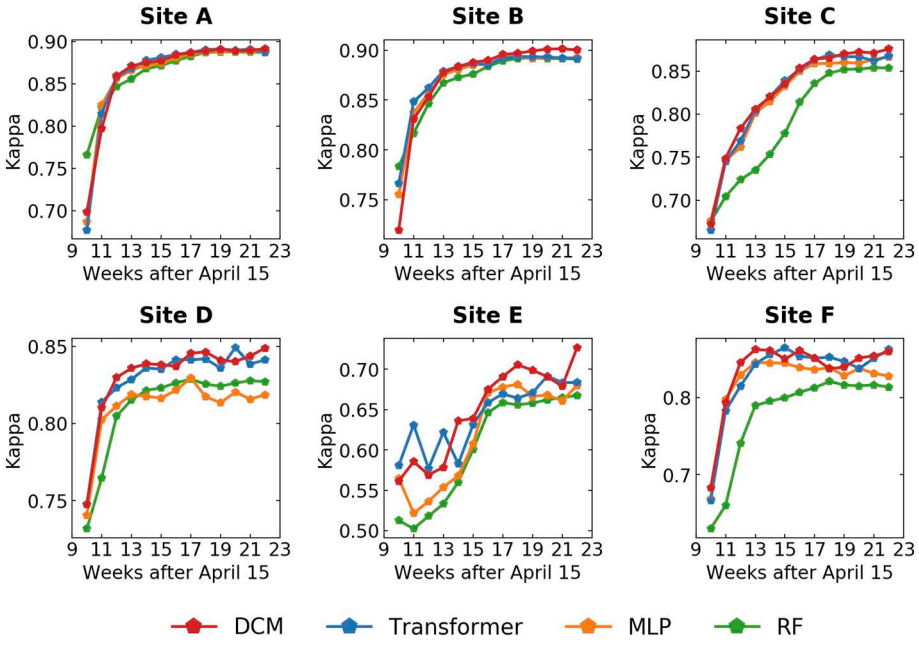


Fig. 11. Cohen's kappa coefficients of DCM, Transformer, RF, and MLP over time for six sites in the in-season classification experiment. The kappa score curves from week 10 (June 24) to week 22 (September 16) are shown here. For each week, the classification model receives the ARD time series consisting of all previous weeks as input to output prediction results at the current week. All kappa scores are achieved by the models, which are trained in Site A from 2015 to 2017 and tested in all sites for the year 2018.

desynchrony of crop progress. Although these sites had similar overall vegetation phenological cycles, crop growth progress at the same week varied spatially, which agreed with the spatial differences of Normalized Difference Vegetation Index (NDVI) temporal profiles in Fig. 4(b). Compared to the base site, there might be leads and lags of phenology phases in transfer sites. Due to the misalignment of crop progress across sites, the generalizable features learned from the base site were not strictly applicable for employing the remote sensing data at the same week to identify crop types in transfer sites. Due to the strong similarity to Site A, Sites B and C suffered less and smaller performance fluctuations than other transfer sites in the in-season classification experiment.

3.4. Understanding the learning process of DeepCropMapping for classification

Increasing the length of seasonal ARD time series reduced the classification uncertainty of DCM based on the analysis of soft outputs (Fig. 12). For each sample, the output module in DCM generated a predicted probability for each category. The predicted probability corresponding to the correct category was selected as classification confidence. Higher confidence implied lower uncertainty and made the classification results more reliable. At the early stage of the growing season, the model was not very reliable with an average confidence lower than 70% at week 2 (April 29) for all the sites. Considering that corn in the study area is mostly planted in late-April and early-May and soybean is mostly planted in mid-May and late-May, some croplands are bare soil at that time. Satellite images could not provide much useful information to distinguish land cover types at the current stage. With the progression of the growing season, DCM achieved better accuracy and produced predictions with higher confidence, indicating the model gradually learned long-term temporal features from multi-temporal remote sensing data with improved confidence. By the end of the growing season, the model reached an average confidence of 91% for the base site and an average confidence higher than 84% for all the transfer sites.

The DCM model gradually learned efficient growth features and dropped trivial noise through the learning procedure reflected by the change of the distribution of classification confidence (Fig. 12). At the early stage, the model obtained low performance, and the confidence concentrated on around 50% with few members at the high and low ends and many in the middle for all the sites except Sites E and F. These two sites had a large number of samples with high confidence at early weeks. It could be explained by their large proportions of “other” category, whose classification required few temporal features. At week 9 (June 17), the majority of samples was correctly classified with high confidence. Some misclassified samples, however, showed very low confidence close to zero. Too short time series could mislead the model to overfit some trivial features and make the wrong prediction since the difference between corn and soybean had not been fully revealed. As more observations included into the classification procedure, these samples were greatly reduced and the model performance was further improved.

The features extracted by the DCM model showed higher separability for different land cover types with the progression of the growing season, as illustrated using Site E as an example (Fig. 13). The hidden features outputted by the AtBiLSTM module in DCM were adopted as the learned temporal features in this study. Samples from the same land cover type were better grouped based on the learned features as the growing season approaches, especially from corn and soybean. The results agreed with the uncertainty analysis in Fig.12, which indicated that the DCM model improved the classification capabilities through the accumulated time-series data. A large number of corn (blue) and soybean (orange) samples located in direct vicinity to each other at the first few weeks when some croplands were bare soil. The spectral differences between crop types were gradually revealed around week 13 (July 15) when corn and soybean grew to about the silking and blooming stage, respectively. The well-trained DCM model captured the critical information and thus extracted efficient features for crop type classification since then. The samples from “other” category (green) were relatively separable from crop samples in general,

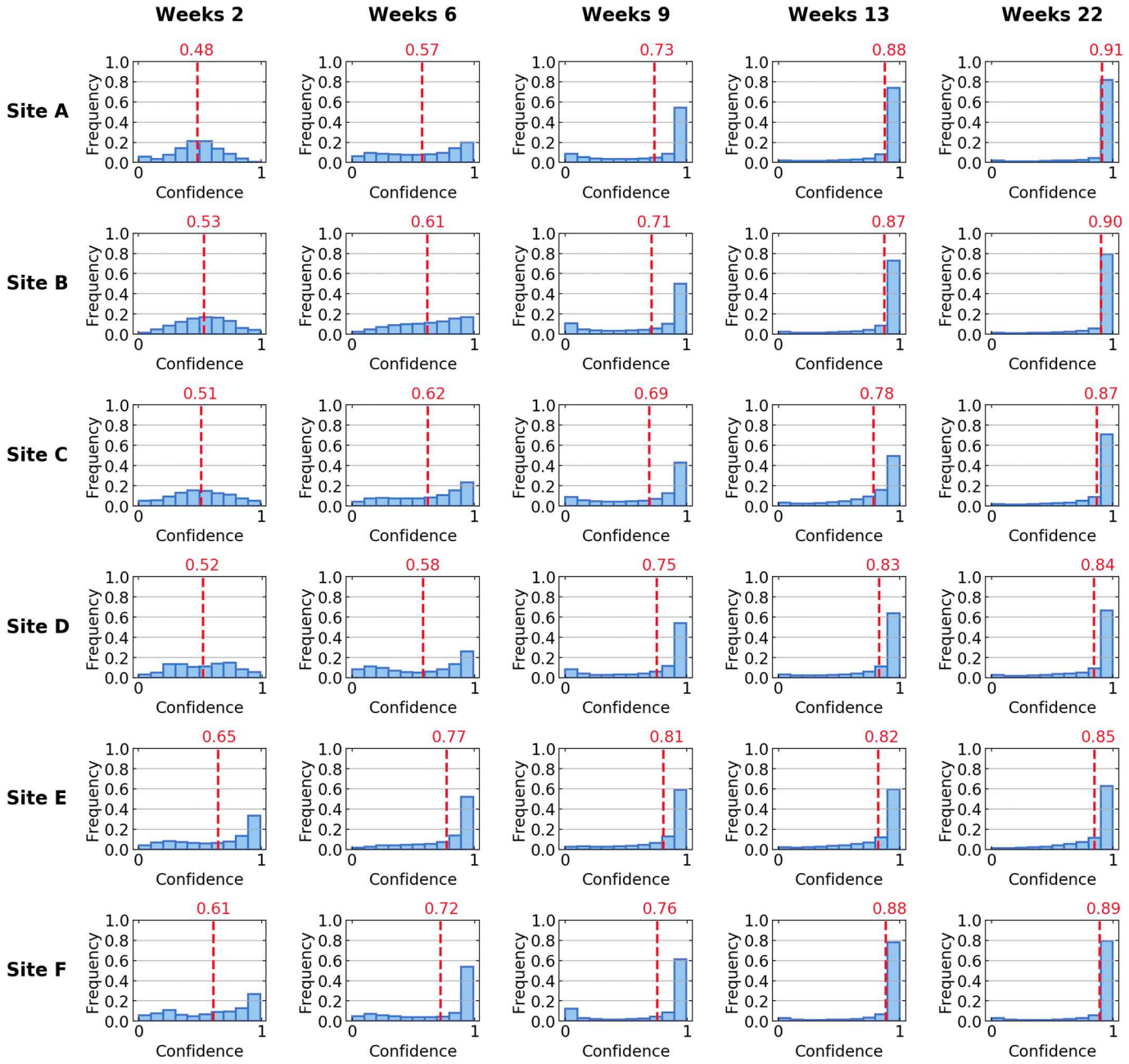


Fig. 12. Temporal changes of the classification confidence distribution by DCM. The DCM model illustrated here is trained in Site A from 2015 to 2017 and tested in all sites for the year 2018. For each sample, the output module in DCM generates a predicted probability for each category, and classification confidence refers to the predicted probability corresponding to the correct category. The frequency showed here is the count divided by the total sample size. The dashed red lines refer to average confidence values. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

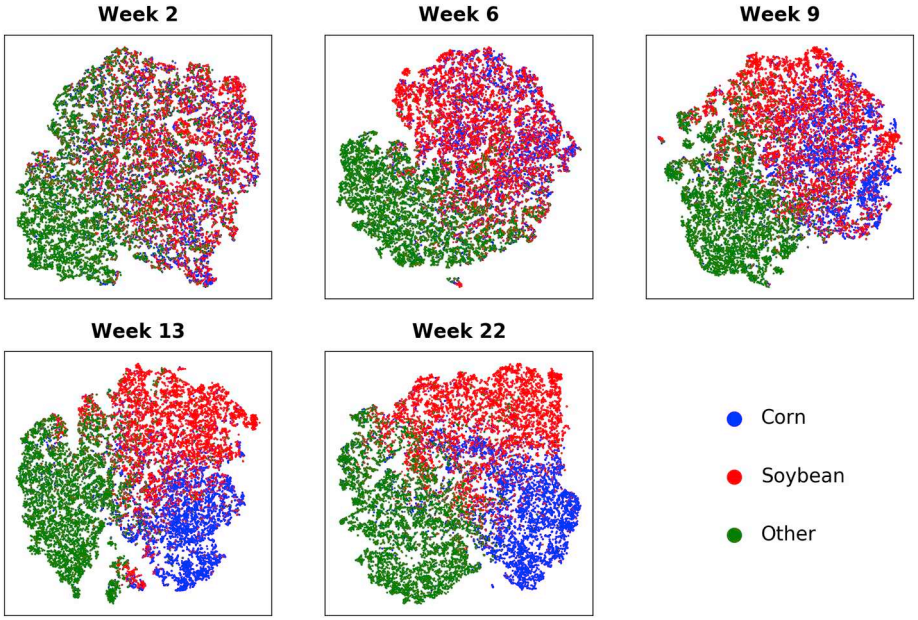


Fig. 13. Visualization of learned features over time based on t-SNE. The learned features of 8000 samples for each land cover type, which are randomly selected from Site E for the year 2018, are nonlinearly projected to two visually interpretable dimensions. Each point stands for one sample.

even at the early stage such as week 6, indicating that discriminating between specific crop types was more challenging and required longer time series than identifying croplands.

4. Future work

Due to the spectral and phenological similarities of corn and soybean, a DCM model was developed to learn accumulated time-series spectral features for an improved generalized classification. Although it achieved competitive performance for crop type classification in this study, the requirement of a complete time series is one major limitation of the current workflow. The threshold of valid observations for filtering out the pixels without sufficient temporal information in preprocessing had a great impact on the proportion of valid pixels as shown in Table S3. The sensitivity of the proportion of valid pixels to the threshold depended on the quality of raw remote sensing data for the specific site and year. Less cloud and shadow and higher revisit frequency improved data quality and reduced sensitivity. Imputing missing data by fitting or interpolation algorithms has been widely applied before time series modeling in previous studies (Inglada et al., 2015; Roy and Yan, 2018; Yan and Roy, 2020). With the rapid growth of available satellite data sources and data fusion techniques, an improved remote sensing time series would less constrain the large-scale application of multi-temporal models like DCM. One possible future work is to investigate the applicability of the combination of data fusion techniques and the DCM architecture. Expert knowledge can also be integrated to optimize input features, such as exploiting NDVI and Enhanced Vegetation Index (EVI) profiles, for facilitating information distillation with limited valid satellite observations.

Further optimization of the proposed DCM architecture will be studied for better performance and interpretation. The combination of convolutional and recurrent neural networks has great potential in spatio-temporal information extraction of remote sensing imagery (Interdonato et al., 2019; Rußwurm and Körner, 2018; Rustowicz et al., 2019). Adding convolutional operations within AtBiLSTM modules or replacing the output module with a fully convolutional network provides possibility to learn diverse feature representation for predicting crop distribution. Similar methods can also contribute to the deeper interpretation of the impact of specific structures in DCM, including bidirectional flows and attention layers. The Transformer architecture facilitates feature extraction from a very long sequence which can exceed thousands of units with efficient parallel computing, which may not well suit agriculture-related remote sensing applications with less than 30 units of time series. Its critical self-attention structures can be further evaluated with an increased frequent remote sensing time series.

Another possible application is to study the model transfer in a large spatial extent. The current study focuses on the spatial transfer between six sites that locate in different U.S. corn maturity zones. Model transfer in larger spatial extent may face the challenges such as the differences in meteorological factors, topographical conditions and crop practice. Future work is needed to transfer the model developed in the U.S. corn belt to areas in China and Brazil using publicly available remote sensing data.

5. Conclusion

A spatially generalizable multi-temporal model named DeepCropMapping (DCM) was developed to learn accumulated time-series spectral features for dynamic corn and soybean mapping in the U.S. corn belt. The DCM model significantly outperformed Transformer, RF, and MLP with a 95% confidence interval in the spatial transfer experiment. Achieving the mean kappa score of 85.8% in base sites and the mean average kappa score of 82.0% in transfer sties at the end of the growing season, DCM yielded the most competitive performance when transferred from the base site to other sites, indicating its capability of extracting generalizable features from remote sensing time series. The results of the in-season classification experiment suggested the performance advantage of DCM at the early stage of the growing season. DCM captured critical information from key growth phases and thus achieved higher accuracy than baseline models after the beginning of July, although all models benefited from the observation increase. To further understand the learning process of DCM, the classification confidence was monitored over time. The results in all sites demonstrated that increasing the length of seasonal remote sensing time series reduced the classification uncertainty of DCM.

This study provided an end-to-end multi-temporal approach for learning generalizable feature representation across regions from satellite observation sequences. Compared to the combination of conventional models and handcrafted phenological features, this approach

learned temporal features efficiently and automatically, depended less on handcraft efforts and suited in-season classification. The spatially generalizable model had great potential in producing near-real-time crop-specific cover maps for agriculture areas that lack reliable ground reference labels.

Code availability

The DCM model is available at https://github.com/Lab-IDEAS/DeepCropMapping.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was partially funded by National Natural Science Foundation of China under Grant Number 31701316 and 41871364 and Zhejiang University.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2020.111946.

References

Sak, H., Senior, A., Beaufays, F., 2014. Long short-term memory recurrent neural network architectures for large scale acoustic modeling. In: Fifteenth Annual Conference of the International Speech Communication Association.

Sundermeyer, M., Schlüter, R., Ney, H., 2012. LSTM neural networks for language modeling. In: Thirteenth Annual Conference of the International Speech Communication Association.

USDA-NASS, 2018. Quick Stats 2.0. USDA-NASS, Washington, DC.

AAFC, 2018. AAFC Annual Crop Inventory – Data Product Specifications (ISO 19131).

Azzari, G., Lobell, D.B., 2017. Landsat-based classification in the cloud: an opportunity for a paradigm shift in land cover monitoring. In: Remote Sensing of Environment, Big Remotely Sensed Data: tools, applications and experiences. 202. pp. 64–74. https://doi.org/10.1016/j.rse.2017.05.025.

Belgiu, M., Drăguţ, L., 2016. Random forest in remote sensing: a review of applications and future directions. ISPRS J. Photogramm. Remote Sens. 114, 24–31.

Boryan, C., Yang, Z., Mueller, R., Craig, M., 2011. Monitoring US agriculture: the US Department of Agriculture, National Agricultural Statistics Service, Cropland Data Layer Program. Geocarto Int. 26, 341–358. https://doi.org/10.1080/10106049.2011.562309.

Bradley, B.A., Mustard, J.F., 2008. Comparison of phenology trends by land cover class: a case study in the Great Basin, USA. Glob. Chang. Biol. 14, 334–346. https://doi.org/10.1111/j.1365-2486.2007.01479.x.

Breiman, L., 1996. Bagging predictors. Mach. Learn. 24, 123–140.

Breiman, L., 2001. Random forests. Mach. Learn. 45, 5–32.

Brown, M.E., de Beurs, K., Vrieling, A., 2010. The response of African land surface phenology to large scale climate oscillations. Remote Sens. Environ. 114, 2286–2296. https://doi.org/10.1016/j.rse.2010.05.005.

Cai, Y., Guan, K., Peng, J., Wang, S., Seifert, C., Wardlow, B., Li, Z., 2018. A high-performance and in-season classification system of field-level crop types using time-series Landsat data and a machine learning approach. Remote Sens. Environ. 210, 35–47. https://doi.org/10.1016/j.rse.2018.02.045.

Carpenter, G.A., Gopal, S., Macomber, S., Martens, S., Woodcock, C.E., Franklin, J., 1999. A neural network method for efficient vegetation mapping. Remote Sens. Environ. 70, 326–338. https://doi.org/10.1016/S0034-4257(99)00051-6.

Chen, Y., Song, X., Wang, S., Huang, J., Mansaray, L.R., 2016. Impacts of spatial heterogeneity on crop area mapping in Canada using MODIS data. ISPRS J. Photogramm. Remote Sens. 119, 451–461. https://doi.org/10.1016/j.isprsjprs.2016.07.007.

Cohen, W.B., Fiorella, M., Gray, J., Helmer, E., Anderson, K., 1998. An efficient and accurate method for mapping forest clearcuts in the Pacific Northwest using Landsat imagery. Photogramm. Eng. Remote. Sens. 64, 293–300.

Dai, Z., Yang, Z., Yang, Y., Carbonell, J., Le, Q.V., Salakhutdinov, R., 2019. Transformer-XL: Attentive Language Models Beyond a Fixed-Length Context. arXiv:1901.02860 [cs, stat].

Devlin, J., Chang, M.-W., Lee, K., Toutanova, K., 2019. Pre-training of Deep Bidirectional Transformers for Language Understanding. arXiv:1810.04805 [cs].

Dixon, B., Candade, N., 2008. Multispectral landuse classification using neural networks and support vector machines: one or the other, or both? Int. J. Remote Sens. 29, 1185–1206. https://doi.org/10.1080/01431160701294661.

Dong, J., Xiao, X., Kou, W., Qin, Y., Zhang, G., Li, L., Jin, C., Zhou, Y., Wang, J., Biradar, C., Liu, J., Moore, B., 2015. Tracking the dynamics of paddy rice planting area in 1986–2010 through time series Landsat images and phenology-based algorithms. Remote Sens. Environ. 160, 99–113. https://doi.org/10.1016/j.rse.2015.01.004.

Edreira, J.I.R., Mayer, L.I., Otegui, M.E., 2014. Heat stress in temperate and tropical maize hybrids: kernel growth, water relations and assimilate availability for grain filling. Field Crop Res. 166, 162–172. https://doi.org/10.1016/j.fcr.2014.06.018.

Garnot, V.S.F., Landrieu, L., Giordano, S., Chehata, N., 2019. Satellite Image Time Series Classification with Pixel-Set Encoders and Temporal Self-Attention. arXiv:1911.07757 [cs].

He, T., Xie, C., Liu, Q., Guan, S., Liu, G., 2019. Evaluation and comparison of random Forest and A-LSTM networks for large-scale winter wheat identification. Remote Sens. 11, 1665. https://doi.org/10.3390/rs11141665.

Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. Neural Comput. 9, 1735–1780.

Huang, B., Zhao, B., Song, Y., 2018. Urban land-use mapping using a deep convolutional neural network with high spatial resolution multispectral remote sensing imagery. Remote Sens. Environ. 214, 73–86. https://doi.org/10.1016/j.rse.2018.04.050.

Inglada, J., Arias, M., Tardy, B., Hagolle, O., Valero, S., Morin, D., Dedieu, G., Sepulcre, G., Bontemps, S., Defourny, P., Koetz, B., 2015. Assessment of an operational system for crop type map production using high temporal and spatial resolution satellite optical imagery. Remote Sens. 7, 12356–12379. https://doi.org/10.3390/rs70912356.

Interdonato, R., Ienco, D., Gaetano, R., Ose, K., 2019. DuPLO: a DUal view point deep learning architecture for time series classification. ISPRS J. Photogramm. Remote Sens. 149, 91–104. https://doi.org/10.1016/j.isprsjprs.2019.01.011.

Jia, X., Khandelwal, A., Nayak, G., Gerber, J., Carlson, K., West, P., Kumar, V., 2017. Incremental dual-memory lstm in land cover prediction. In: Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 867–876.

Khatami, R., Mountrakis, G., Stehman, S.V., 2016. A meta-analysis of remote sensing research on supervised pixel-based land-cover image classification processes: general guidelines for practitioners and future research. Remote Sens. Environ. 177, 89–100. https://doi.org/10.1016/j.rse.2016.02.028.

King, L., Adusei, B., Stehman, S.V., Potapov, P.V., Song, X.-P., Krylov, A., Di Bella, C., Loveland, T.R., Johnson, D.M., Hansen, M.C., 2017. A multi-resolution approach to national-scale cultivated area estimation of soybean. Remote Sens. Environ. 195, 13–29. https://doi.org/10.1016/j.rse.2017.03.047.

Kingma, D.P., Ba, J., 2014. Adam: A method for stochastic optimization. arXiv preprint arXiv:1412.6980.

Kussul, N., Lavreniuk, M., Skakun, S., Shelestov, A., 2017. Deep learning classification of land cover and crop types using remote sensing data. IEEE Geosci. Remote Sens. Lett. 14, 778–782. https://doi.org/10.1109/LGRS.2017.2681128.

LeCun, Y., Bengio, Y., Hinton, G., 2015. Deep learning. Nature 521, 436–444. https://doi.org/10.1038/nature14539.

Lobell, D.B., Azzari, G., 2017. Satellite detection of rising maize yield heterogeneity in the US Midwest. Environ. Res. Lett. 12, 014014.

Löw, F., Michel, U., Dech, S., Conrad, C., 2013. Impact of feature selection on the accuracy and spatial uncertainty of per-field crop classification using support vector machines. ISPRS J. Photogramm. Remote Sens. 85, 102–119. https://doi.org/10.1016/j.isprsjprs.2013.08.007.

Lyu, H., Lu, H., Mou, L., 2016. Learning a transferable change rule from a recurrent neural network for land cover change detection. Remote Sens. 8, 506. https://doi.org/10.3390/rs8060506.

van der Maaten, L., Hinton, G., 2008. Visualizing Data using t-SNE. J. Mach. Learn. Res. 9, 2579–2605.

Marcos, D., Volpi, M., Kellenberger, B., Tuia, D., 2018. Land cover mapping at very high resolution with rotation equivariant CNNs: towards small yet accurate models. ISPRS J. Photogrammet. Remote Sensing 145, 96–107. https://doi.org/10.1016/j.isprsjprs.2018.01.021.

Mas, J.F., Flores, J.J., 2008. The application of artificial neural networks to the analysis of remotely sensed data. Int. J. Remote Sens. 29, 617–663. https://doi.org/10.1080/01431160701352154.

Massey, R., Sankey, T.T., Congalton, R.G., Yadav, K., Thenkabail, P.S., Ozdogan, M., Sánchez Meador, A.J., 2017. MODIS phenology-derived, multi-year distribution of conterminous U.S. crop types. Remote Sens. Environ. 198, 490–503. https://doi.org/10.1016/j.rse.2017.06.033.

Mou, L., Bruzzone, L., Zhu, X.X., 2019. Learning spectral-spatial-temporal features via a recurrent convolutional neural network for change detection in multispectral imagery. IEEE Trans. Geosci. Remote Sens. 57, 924–935. https://doi.org/10.1109/TGRS.2018.2863224.

Nelson, G.C., Rosegrant, M.W., Koo, J., Robertson, R., Sulser, T., Zhu, T., Ringler, C., Msangi, S., Palazzo, A., Batka, M., 2009. Climate Change: Impact on Agriculture and Costs of Adaptation. (Intl Food Policy Res Inst).

Pax-Lenney, M., Woodcock, C.E., Macomber, S.A., Gopal, S., Song, C., 2001. Forest mapping with a generalized classifier and Landsat TM data. Remote Sens. Environ. 77, 241–250. https://doi.org/10.1016/S0034-4257(01)00208-5.

Pelletier, C., Valero, S., Inglada, J., Champion, N., Dedieu, G., 2016. Assessing the robustness of random forests to map land cover with high resolution satellite image time series over large areas. Remote Sens. Environ. 187, 156–168. https://doi.org/10.1016/j.rse.2016.10.010.

Rodriguez-Galiano, V.F., Ghimire, B., Rogan, J., Chica-Olmo, M., Rigol-Sanchez, J.P., 2012. An assessment of the effectiveness of a random forest classifier for land-cover classification. ISPRS J. Photogramm. Remote Sens. 67, 93–104.

Roy, D.P., Yan, L., 2018. Robust Landsat-based crop time series modelling. Remote Sens. Environ. https://doi.org/10.1016/j.rse.2018.06.038.

Rußwurm, M., Körner, M., 2017. Temporal Vegetation Modelling Using Long Short-Term Memory Networks for Crop Identification from Medium-Resolution Multi-spectral Satellite Images. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). Presented at the 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), pp. 1496–1504. https://doi.org/10.1109/CVPRW.2017.193.

Rußwurm, M., Körner, M., 2018. Multi-temporal land cover classification with sequential recurrent encoders. ISPRS Int. J. Geo Inf. 7, 129. https://doi.org/10.3390/ijgi7040129.

Rußwurm, M., Körner, M., 2019. Self-Attention for Raw Optical Satellite Time Series Classification. arXiv:1910.10536 [cs, eess, stat].

Rustowicz, R., Cheong, R., Wang, L., Ermon, S., Burke, M., Lobell, D., 2019. Semantic Segmentation of Crop Type in Africa: A Novel Dataset and Analysis of Deep Learning Methods. In: Presented at the Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, pp. 75–82.

Sakamoto, T., Van Nguyen, N., Ohno, H., Ishitsuka, N., Yokozawa, M., 2006. Spatio–temporal distribution of rice phenology and cropping systems in the Mekong Delta with special reference to the seasonal water flow of the Mekong and Bassac rivers. Remote Sens. Environ. 100, 1–16. https://doi.org/10.1016/j.rse.2005.09.007.

Shi, D., Yang, X., 2016. An assessment of algorithmic parameters affecting image classification accuracy by random forests. Photogramm. Eng. Remote Sens. 82, 407–417. https://doi.org/10.1016/S0099-1112(16)82032-7.

Siachalou, S., Mallinis, G., Tsakiri-Strati, M., 2015. A hidden Markov models approach for crop classification: linking crop phenology to time series of multi-sensor remote sensing data. Remote Sens. 7, 3633–3650. https://doi.org/10.3390/rs70403633.

Simonneaux, V., Francois, P., 2003. Identifying main crop classes in an irrigated area using high resolution image time series. In: in: IGARSS 2003. 2003 IEEE International Geoscience and Remote Sensing Symposium. Proceedings (IEEE Cat. No. 03CH37477), pp. 252–254.

Simonneaux, V., Duchemin, B., Helson, D., Er-Raki, S., Olioso, A., Chehbouni, A.G., 2008. The use of high-resolution image time series for crop classification and evapotranspiration estimate over an irrigated area in Central Morocco. Int. J. Remote Sens. 29, 95–116. https://doi.org/10.1080/01431160701250390.

Song, X.-P., Potapov, P.V., Krylov, A., King, L., Di Bella, C.M., Hudson, A., Khan, A., Adusei, B., Stehman, S.V., Hansen, M.C., 2017. National-scale soybean mapping and area estimation in the United States using medium resolution satellite imagery and field survey. Remote Sens. Environ. 190, 383–395. https://doi.org/10.1016/j.rse.2017.01.008.

Soudani, K., le Maire, G., Dufrêne, E., François, C., Delpierre, N., Ulrich, E., Cecchini, S., 2008. Evaluation of the onset of green-up in temperate deciduous broadleaf forests derived from moderate resolution imaging Spectroradiometer (MODIS) data. In: Remote Sensing of Environment, Earth Observations for Terrestrial Biodiversity and Ecosystems Special Issue. 112. pp. 2643–2655. https://doi.org/10.1016/j.rse.2007.12.004.

Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R., 2014. Dropout: a simple way to prevent neural networks from overfitting. J. Mach. Learn. Res. 15, 1929–1958.

Thenkabail, P.S., Knox, J.W., Ozdogan, M., Gumma, M.K., Congalton, R.G., Wu, Z., Milesi, C., Finkral, A., Marshall, M., Mariotto, I., et al., 2012. Assessing future risks to agricultural productivity, water resources and food security: how can remote sensing help? In: PE&RS, Photogrammetric Engineering & Remote Sensing. 78. pp. 773–782.

USDA National Agricultural Statistics Service Cropland Data Layer, 2018. Published Crop-Specific Data Layer.

USGS ARD, 2018. U.S. Geological Survey, Landsat Analysis Ready Data (ARD).

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need. In: Guyon, I., Luxburg, U.V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., Garnett, R. (Eds.), Advances in Neural Information Processing Systems 30. Curran Associates, Inc., pp. 5998–6008.

Vermeulen, S.J., Campbell, B.M., Ingram, J.S., 2012. Climate change and food systems. Annu. Rev. Environ. Resour. 37, 195–222.

Walker, J.J., de Beurs, K.M., Henebry, G.M., 2015. Land surface phenology along urban to rural gradients in the U.S. Great Plains. Remote Sens. Environ. 165, 42–52. https://doi.org/10.1016/j.rse.2015.04.019.

Wang, S., Azzari, G., Lobell, D.B., 2019. Crop type mapping without field-level labels: random forest transfer and unsupervised clustering techniques. Remote Sens. Environ. 222, 303–317. https://doi.org/10.1016/j.rse.2018.12.026.

Werbos, P.J., 1990. Backpropagation through time: what it does and how to do it. Proc. IEEE 78, 1550–1560.

Woodcock, C.E., Macomber, S.A., Pax-Lenney, M., Cohen, W.B., 2001. Monitoring large areas for forest change using Landsat: generalization across space, time and Landsat sensors. Remote sensing of environment. Landsat 7 (78), 194–203. https://doi.org/10.1016/S0034-4257(01)00259-0.

Yan, L., Roy, D.P., 2020. Spatially and temporally complete Landsat reflectance time series modelling: the fill-and-fit approach. Remote Sens. Environ. 241, 111718. https://doi.org/10.1016/j.rse.2020.111718.

Zhang, G., Xiao, X., Dong, J., Kou, W., Jin, C., Qin, Y., Zhou, Y., Wang, J., Menarguez, M.A., Biradar, C., 2015. Mapping paddy rice planting areas through time series analysis of MODIS land surface temperature and vegetation index data. ISPRS J. Photogramm. Remote Sens. 106, 157–171. https://doi.org/10.1016/j.isprsjprs.2015.05.011.

Zhong, L., Gong, P., Biging, G.S., 2014. Efficient corn and soybean mapping with temporal extendability: a multi-year experiment using Landsat imagery. Remote Sens. Environ. 140, 1–13. https://doi.org/10.1016/j.rse.2013.08.023.

Zhong, L., Hu, L., Zhou, H., 2019a. Deep learning based multi-temporal crop classification. Remote Sens. Environ. 221, 430–443. https://doi.org/10.1016/j.rse.2018.11.032.

Zhong, L., Hu, L., Zhou, H., Tao, X., 2019b. Deep learning based winter wheat mapping using statistical data as ground references in Kansas and northern Texas, US. Remote Sens. Environ. 233, 111411. https://doi.org/10.1016/j.rse.2019.111411.

Zhou, P., Shi, W., Tian, J., Qi, Z., Li, B., Hao, H., Xu, B., 2016. Attention-based bidirectional long short-term memory networks for relation classification. In: in: Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), pp. 207–212.

Zipper, S.C., Qiu, J., Kucharik, C.J., 2016. Drought effects on US maize and soybean production: spatiotemporal patterns and historical changes. Environ. Res. Lett. 11, 094021. https://doi.org/10.1088/1748-9326/11/9/094021.